SURFACE CHEMISTRY

1. When a chalk stick is dipped in Ink solution then the ink particles are.
(1) Adsorbed
(2) Absorbed
(3) Sorption takes place
(4) Precipitated

2. In physical adsorption, when rate of adsorption and rate of desorption are equal then
(1) $\Delta H = 0$ (2) $T\Delta S = 0$
(3) $\Delta G = 0$ (4) All the above

3. In Freundlich Adsorption isotherm equation when $\frac{1}{n} = 0$
(1) Adsorption is independent of pressure
(2) Adsorption varies directly with pressure
(3) Adsorption varies inversely with pressure
(4) Depends on pressure value

4. What is the probable range of 1/n in Freundlich Adsorption isotherm equation.
(1) 0 – 1 (2) 0 – 0.5
(3) 0.5 – 1 (4) 0.1 – 0.5

5. Freundlich Adsorption isotherm fails at
(1) Low pressure
(2) High pressure
(3) Low temperature
(4) Low temperature & low pressure

6. When a charcoal is put into acetic acid solution
(1) concentration increases
(2) concentration decreases
(3) concentration remains same
(4) Acetic acid precipitates

7. Precipitate of $Mg(OH)_2$ attains blue colour in presence of magneson reagent due to
(1) Absorption (2) Adsorption
(3) Sorption (4) desorption

8. In Freundlich equation $\frac{x}{m} = KC^{1/n}$, C is
(1) Initial concentration
(2) Equilibrium concentration
(3) concentration of Adsorbent
(4) Average concentration

9. For Hydrogenation reactions maximum catalytic activity is shown by
(1) Group 5 – 11 elements
(2) Group 5 – 9 elements
(3) Group 7 – 9 elements
(4) Group 9 – 11 elements

10. If a micellar solution of soap is diluted, then
(1) More micelles are formed
(2) lytotropic mesomorphs are formed
(3) Soap precipitates out
(4) Micelles break into ions

11. Which of the following can be used to speed up the ultrafiltration process.
(a) Pressure
(b) Suction
(c) Increase in pore size
(d) Adding dispersion medium
(1) a, b (2) a, c
(3) a, b, c (4) a, c, d

12. Finest gold sol has colour
(1) Red (2) blue
(3) Purple (4) violet

13. On increasing size of colloidal particle of gold from smallest one, the colour shifts from
(1) Red to Golden
(2) blue to Red
(3) Remains same
(4) Darks Same

14. Charge on sol particles is
(A) Due to electron capture by sol particles
(B) Due to preferential adsorption of ions
(C) Due to formulation of electrical double layer
(1) a and b (2) a and c
(3) a, b and c (4) b and c

15. Positively charged sols is/are–
(1) $Al_2O_3 . xH_2O$ and $CrO_3 . x H_2O$
(2) Haemoglobin
(3) Methylene blue sol.
(4) All

16. Tyndal effect is observed when–

(A) Diameter of dispersed particles is not much smaller than the wavelength of light used

(B) The refractive indices of the dispersed phase and the dispersion medium differ greatly in magnitude

(C) Diameter of dispersed particles is much larger than the wavelength of the light used

(1) a and b (2) a and c

(3) b and c (4) a, b and c

17. CMC [critical miscelle concentration] for soaps is–

(1) 10^{-4} to 10^{-3} mol/lit.

(2) 10^{-6} to 10^{-2} mol/lit.

(3) 10^{-10} to 10^{-7} mol/lit.

(4) 10^{-12} to 10^{-5} mol/lit.

18. Optimum temperature range for enzymatic activity is–

(1) 298 K to 310 K (2) 273 K to 310 K

(3) 273 K to 298 K (4) 298 K to 330 K

19. The volume of nitrogen gas at 0°C and 1.013 bar required to cover a sample of silica gel with unimolecular layer is 129 cm^3/gm of gel. Then the surface area per gm of the gel will be if each nitrogen molecule occupies 16.2×10^{-20} m^2.

(1) 568 m^2 gm^{-1}

(2) 5.68 m^2 gm^{-1}

(3) 5680 $m^2 gm^{-1}$

(4) 5680 cm^2 gm^{-1}

20. The arsenious sulphide sol has negative charge. The maximum coagulating power for precipitating it is of–

(1) 0.1 N $Zn(NO_3)_2$

(2) 0.1 N Na_3PO_4

(3) 0.1 N $ZnSO_4$

(4) 0.1 N $AlCl_3$